

Pollution control for a spatially structured economic growth system

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In this paper investigations by the same authors on environmental issues concerning the control of the pollution produced by human activities have been extended to include costs related to environmental interventions.

The proposed model consists of a spatially structured dynamic model which takes into account the level of pollution induced by human activities and possible environmental interventions, including taxation. An optimal harvesting control problem has been discussed, with an objective function composed of four terms, namely the intertemporal utility of the decision maker, the space-time average of the level of pollution in the habitat, the disutility due to the imposition of taxation and the cost of environmental interventions.

A specific novelty in the model proposed here is the localization of the possible interventions to a subregion of the whole habitat.

Computational experiments have been carried out to exemplify the outcomes of the proposed model.

Key words: geographical economics, environmental pollution, reaction-diffusion systems, non-local interactions, optimal control, regional control, generalized production functions

1. Introduction

In this paper we introduce an environmental issue concerning the control of the pollution produced by human activities.

Standard macroeconomics and environmental economics have been two completely independent research areas for a long time. The seminal paper by Gross-

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man and Kruger [21] has stimulated considerable interest in the relationship between environmental pollution and economic development. The role of taxation for controlling pollution is the key issue of the report [24] (see also references therein) issued by the International Monetary Fund. In view of these policy developments, some recent works tend to develop a global theory combining the above mentioned two branches of literature (see [14] and [11]).

Here we present an extension of our own results concerning the impact of pollution diffusion on the economic and environmental joint dynamics, and consider the optimal control problem of the relevant system subject to capital consumption, to taxation based on the amount of produced pollution, and possible environmental interventions.

A key issue of this paper is the localization of possible interventions to a subregion of the whole habitat, in order to reduce costs of the intervention itself and those due to logistic peculiarities (see [6], and references therein).

In [2] and [3] the authors introduced a reaction-diffusion system modelling the impact of environmental pollution on an economic growth system. In particular, in [2] the authors analyzed the large-time behavior of a spatially structured economic growth model coupling physical capital accumulation and pollution diffusion: this model extends other results in the literature along different directions. Furthermore, a negative feedback to the production function was added, in order to describe the (negative) impact of pollution on capital accumulation. In [5] the impact of taxation on production was faced; an optimal control problem was analyzed with a cost functional including taxation as opposed to economic growth. This includes taxation on all activities that may have a huge impact on pollution emissions.

In the literature other contributions which explore the spatial dimension in environmental and resource economics can be found in [7, 15, 16, 29]. Of interest it might be also [10], concerning the optimal control of age dependent systems.

The inclusion of a spatial dimension in economic analysis has recently increased relevance and interest. The first studies in economic geography go back to Beckerman [9] and Puu [25], who study regional problems based simply on flow equations. Starting from these works, a new economic geography arises adopting general equilibrium models to analyze the peculiarities of local and global markets, and the mobility of production factors (see [19, 22]). More recently, this geographical approach has been introduced in economic growth models to study the implications between accumulation and diffusion of capital on economic dynamics (see [12, 13, 17]). The Solow model [27] with a continuous spatial dimension has been extensively studied. In [17] the authors analyze problems of convergence across regions when capital is mobile, while [13] considers the case in which both capital and labor are mobile. The Ramsey model [26] has been

extended to a spatial dimension in [13] and [12], respectively in average and total utilitarianism version. In [18], alongside the classical Cobb–Douglas production function, the authors have proposed a larger class of convex–concave production functions, thus allowing a larger variety of dynamical scenarios, including the existence of poverty traps. For a recent review see [8].

In addition to the impact of taxation on the production as emerged in [5], in this paper we consider the role of additional costs due to possible environmental interventions.

For reaction models with nonlocal reaction terms in mathematical biology the reader may refer to [6, 20, 28].

As for a large amount of mathematical models, it is rather hard to identify real parameters; scope of our presentation is to evidence, within a conceptual setting, as already proposed in existing literature, that direct reduction of pollution has a clear positive impact on both environmental quality and production. The hope is that the outcomes emerging from this manuscript can inspire decision makers/public environmental agencies to implement adequate environmental practices. As an additional remark, the authors hope that interested readers in environmental issues may design suitable experiments inspired by our model, so to possibly identify real parameters, leading to more realistic computational experiments.

In Section 3 the optimal control problem is presented, by acting on both consumption, taxation, and environmental interventions. Since the search of the optimal control is not an easy analytical task, we have proposed a computational algorithm for the search by the gradient method. To this aim, in the same Section 3 the evaluation of the directional derivative of the cost functional with respect to the three controls has been presented. In Section 4 a numerical algorithm that allows to solve the system (4), (8), (9) of four forward-backward reaction-diffusion equations, subject to the relevant initial/final and boundary conditions, has been reported, together with a set of computational experiments (see Section 5). A discussion on the outcomes of the computational experiments is presented in Section 6.

2. A spatially structured model

By following our previous literature on the subject (see [5], and references therein), we shall consider the following spatially structured model. As far as the evolution of the stock capital is concerned, let $k(x, t)$ denote the capital stock held by the representative household located at $x \in \Omega$ (a nonempty bounded domain in $\mathbb{R}^n$, $n = 1, 2$ with a sufficiently smooth boundary), at time $t \geq 0$. The time

evolution of the physical capital is modelled by

$$\frac{\partial k}{\partial t}(x, t) = d_1 \Delta k(x, t) + A(x, t)f(k(x, t)) - \delta_1 k(x, t) - c(x, t)k(x, t), \quad (1)$$

for $(x, t) \in \Omega \times (0, +\infty)$.

Here $A(x, t)$ denotes the technological progress available at $x \in \Omega$, and time $t \geq 0$. The parameter δ_1 takes into account the depreciation of the capital. $c(x, t)$ describes the level of consumption per unit of physical capital $k(x, t)$; the quantity $(1 - c(x, t))k(x, t)$ describes the saved capital after consumption.

Equation (1) is complemented by an initial capital distribution, $k(x, 0)$, and suitable boundary conditions. If we assume that there is no capital flow through the boundary $\partial\Omega$, we impose homogeneous Neumann boundary conditions, i.e.

$$\frac{\partial k}{\partial n}(x, t) = 0, \quad x \in \partial\Omega, \quad t > 0.$$

The production function f can be taken of the following form (see [18], and references therein)

$$f(r) = \frac{\alpha_1 r^\gamma}{1 + \alpha_2 r^\gamma},$$

where $\alpha_1 \in (0, +\infty)$, $\alpha_2 \in [0, +\infty)$, $\gamma \in [1, +\infty)$.

As far as pollution diffusion is concerned, if we denote by $p(x, t)$ the pollution stock faced by a representative household located at $x \in \Omega$, at time $t \geq 0$, we assume the following model

$$\frac{\partial p}{\partial t}(x, t) = d_2 \Delta p(x, t) + \theta \int_{\Omega} f(k(x', t))\varphi(x', x) dx' - \delta_2 p(x, t). \quad (2)$$

The first term, including the Laplace operator, may model the random dispersal of the pollutant in the relevant habitat. Via the integral term in Equation (2), we may see the impact of the economy on pollution production. At any point $x \in \Omega$ pollution is generated due to the production $f(k(x, t))$. In our model we have supposed that any point $x' \in \Omega$ may have an impact on enhancing pollution at any other point $x \in \Omega$, by any other possible mechanism than random dispersal already described by the Laplace operator. Such mechanisms are described by a kernel φ which satisfies the following hypotheses: $\varphi \in L^\infty(\Omega \times \Omega)$, and $\varphi(x', x) \geq 0$ a.e. $(x', x) \in \Omega \times \Omega$. As a limiting case one may reduce the kernel φ to a Dirac Delta function, to mean the impact of production on pollution at the same point x . The parameter θ takes into account the “efficiency” of the economic system with respect to pollution production. The parameter δ_2 accounts for the natural decay of pollution.

Equation (2) has to be complemented by suitable initial and boundary conditions.

We have required that the model takes into account the negative feedback due to the impact of pollution on production. The quantity $g(p)$ can be interpreted as a factor inhibiting the performance of the economy, due to a possible negative impact of the pollution on the production process, and/or to a self regulation of the rate of production for reducing pollution taxes (see our papers [2–5], and [23, 30]); an interesting general discussion on this topic can be found in [1].

As a consequence in [5] we modified Equation (1) as follows

$$\begin{aligned} \frac{\partial k}{\partial t}(x, t) = & d_1 \Delta k(x, t) \\ & + \frac{A(x, t)}{g(p(x, t))} f(k(x, t)) - \delta_1 k(x, t) - c(x, t)k(x, t). \end{aligned} \quad (3)$$

All parameters, $A, d_1, d_2, \delta_1, \delta_2$, which characterize a specific economic/ecological system are assumed strictly positive.

The function $g: [0, +\infty) \rightarrow (0, +\infty)$ models the negative impact of pollution on the production; usual assumptions on g are

- i) g is continuously differentiable and monotonically increasing;
- ii) $g(0) = 1$ and $\lim_{p \rightarrow +\infty} g(p) = +\infty$.

As a typical choice for the function g , later we shall adopt the expression

$$g(p) = 1 + \chi p^2, \quad p \geq 0.$$

In the sequel it will be kept $\chi = 1$, as a typographical simplification.

3. An optimal control problem

We plan to study our eco-economic system on a finite time horizon $[0, T]$. We shall denote by

$$Q_T = \Omega \times (0, T); \quad \Sigma_T = \partial\Omega \times (0, T).$$

In order to control pollution emissions by production processes, local authorities may impose taxes proportional to the level of production. A possible approach might be to impose taxes related to interventions on pollution abatement policies implemented only in the areas of interest, which we have indicate by a subregion $\omega \subset \Omega$ of the whole habitat. We shall denote by $\tau(x, t)$ the environmental taxation rate. By taking all this into account, we need to modify System (3)–(2) as

follows (see [5, 14]):

$$\left\{ \begin{array}{l} \frac{\partial k}{\partial t}(x, t) = d_1 \Delta k(x, t) + \frac{A(x, t)}{1 + p(x, t)^2} f((1 - \mathbb{I}_\omega(x) \tau(x, t)) k(x, t)) \\ \quad - \delta_1 k(x, t) - c(x, t) k(x, t), \\ \frac{\partial p}{\partial t}(x, t) = d_2 \Delta p(x, t) + \theta \int_{\Omega} f((1 - \mathbb{I}_\omega(x') \tau(x', t)) k(x', t)) \varphi(x', x) dx' \\ \quad - \delta_2 p(x, t) - \mathbb{I}_\omega(x) \xi(x, t) p(x, t), \end{array} \right. \quad (4)$$

for $(x, t) \in Q_T$, where ω is a nontrivial open subset of Ω . System (4) shall be complemented by initial and boundary conditions to be precised later.

The term $(1 - \tau(x, t))k(x, t)$ accounts for the amount of the physical capital left to be used in the gross domestic production. An increase in the level of taxation allows the implementation of practices leading to an abatement of the level of pollution emissions, so that we may interpret it as a “green” taxation policy. By $\xi(x, t)$ we have denoted the rate of pollution abatement by suitable environmental interventions.

In this case the optimal control problem would be

$$\begin{aligned} \text{Maximize}_{\substack{(c, \tau, \xi) \in \tilde{U}, \\ \omega \subset \Omega}} \left\{ \int_0^T \int_{\Omega} c(x, t) k(x, t) dx dt - \beta_0 \int_0^T \int_{\Omega} p(x, t) dx dt \right. \\ \quad - \beta_1 \int_0^T \int_{\omega} \tau(x, t) k(x, t) dx dt \\ \quad \left. - \beta_2 \int_0^T \int_{\omega} \xi(x, t) p(x, t) dx dt \right\}, \end{aligned}$$

subject to System (4) and the initial and boundary conditions given below.

It is clear that the solutions of the above problems would depend upon the geographical distribution of the potentially polluting production activities; in this respect we may expect a crucial role played by the functional parameter $A(x, t)$. This fact would confirm the relevance of spatially structured models in geographical economics.

For simplicity we may consider homogeneous Neumann boundary conditions

$$\frac{\partial k}{\partial n}(x, t) = \frac{\partial p}{\partial n}(x, t) = 0, \quad (x, t) \in \Sigma_T, \quad (5)$$

and initial conditions

$$k(x, 0) = k_0(x), \quad p(x, 0) = p_0(x), \quad x \in \Omega, \quad (6)$$

where $T > 0$ is fixed and $k_0, p_0 \in L^\infty(\Omega)$, $k_0(x) \geq 0$, $p_0(x) \geq 0$ a.e. $x \in \Omega$.

We impose the constraints

$$1 - c(x, t) - \tau(x, t) \geq s, \quad c(x, t), \tau(x, t) \geq 0, \quad \text{a.e. } (x, t) \in Q_T, \quad (7)$$

where $s > 0$ is the saving factor. In addition, we may assume that $0 \leq \xi(x, t) \leq L$, a.e. $x \in \omega, t \in (0, T)$, for a given constant L .

If we denote by $U = \{(c, \tau) \in L^\infty(Q_T) \times L^\infty(Q_T) : (c, \tau) \text{ satisfies (7)}\}$, the set of controls is $\bar{U} := U \times \{\xi \in L^\infty(\omega \times (0, T)) : 0 \leq \xi(x, t) \leq L, \text{ a.e. } (x, t) \in \omega \times (0, T)\}$ and $\beta_0, \beta_1, \beta_2$ are positive constants; (k, p) is the solution to (4)–(6) corresponding to Q_T and Σ_T . The objective function is composed of four terms that we hereby describe:

T1: the term $\int_0^T \int_\Omega c(x, t) k(x, t) dx dt$, to be maximized, describes the intertemporal utility of the decision maker,

T2: the second term $\beta_0 \int_0^T \int_\Omega p(x, t) dx dt$, to be minimized, represents an average of the total level of pollution across space and time,

T3: the third term $\beta_1 \int_0^T \int_\Omega \tau(x, t) k(x, t) dx dt$, to be minimized, takes into account the disutility due to imposition of taxation.

T4: the fourth term $\beta_2 \int_0^T \int_\Omega \xi(x, t) p(x, t) dx dt$, to be minimized, takes into account the costs due to environmental interventions.

The following modifications occur when we deal with an optimal control problem related to a system of two parabolic equations (with linear diffusions), like ours, with respect to the related optimal control problem associated to the corresponding ordinary differential system: due to the presence of $A_1 k$ in the

first equation of the primal system (the equation satisfied by k) and to $A_2 p$ in the second equation of the primal system (the equation satisfied by p), where A_1 and A_2 are linear elliptic operators with homogeneous boundary conditions, we have to add $-A_1^* \lambda_k$ in the right-hand side of the first costate equation and $-A_2^* \lambda_p$ in the right-hand side of the second co-state equation. Here A_j^* is the adjoint of A_j , $j \in \{1, 2\}$. Since in our case $A_j = d_j \Delta$ and the boundary conditions are of Neumann type we get the presence of $-d_1 \Delta \lambda_k$ in the right-hand side of the first co-state equation, of $-d_2 \Delta \lambda_p$ in the right-hand side of the second co-state equation and of the Neumann boundary conditions.

As a consequence, the costate equations for the previous optimal control problem are:

$$\begin{aligned} \frac{\partial \lambda_k}{\partial t}(x, t) = & -d_1 \Delta \lambda_k(x, t) - c(x, t) + \beta_1 \mathbb{I}_\omega(x) \tau(x, t) \\ & - \lambda_k(x, t) \left[\frac{A(x, t)(1 - \mathbb{I}_\omega(x) \tau(x, t))}{1 + p(x, t)^2} \right. \\ & \times f'((1 - \mathbb{I}_\omega(x) \tau(x, t))k(x, t)) - \delta_1 - c(x, t) \Big] \\ & - (1 - \mathbb{I}_\omega(x) \tau(x, t)) \\ & \times f'((1 - \mathbb{I}_\omega(x) \tau(x, t))k(x, t)) \theta \int_{\Omega} \lambda_p(x', t) \varphi(x, x') dx', \quad (8) \end{aligned}$$

$$\begin{aligned} \frac{\partial \lambda_p}{\partial t}(x, t) = & -d_2 \Delta \lambda_p(x, t) + \beta_0 + \beta_2 \mathbb{I}_\omega(x) \xi(x, t) \\ & + \lambda_k(x, t) \frac{2A(x, t)p(x, t)}{(1 + p(x, t)^2)^2} f'((1 - \mathbb{I}_\omega(x) \tau(x, t))k(x, t)) \\ & + \delta_2 \lambda_p(x, t) + \mathbb{I}_\omega(x) \xi(x, t) \lambda_p(x, t) \quad (9) \end{aligned}$$

together with homogeneous Neumann boundary conditions and zero final data.

As in [4] it follows the well posedness of the initial/boundary problem for System (4). Moreover it follows in the same manner that there exists at least one optimal control (c^*, τ^*, ξ^*) .

Let us now proceed with the construction of a computational algorithm for the search of an optimal control by the gradient method.

For the sake of clarity we shall use the notation k^u , p^u , λ_k^u , λ_p^u for k , p , λ_k , λ_p , where $u = (x, \tau, \xi) \in \tilde{U}$ when some danger of confusion occurs. Denote by

$$\begin{aligned} I(u) = & \int_0^T \int_{\Omega} c(x, t) k^u(x, t) dx dt - \beta_0 \int_0^T \int_{\Omega} p^u(x, t) dx dt \\ & - \beta_1 \int_0^T \int_{\Omega} \mathbb{I}_{\omega}(x) \tau(x, t) k^u(x, t) dx dt - \beta_2 \int_0^T \int_{\Omega} \mathbb{I}_{\omega}(x) \xi(x, t) p^u(x, t) dx dt \end{aligned}$$

the value of the cost functional corresponding to $u = (c, \tau, \xi)$.

Consider an arbitrary $v = (v_1, v_2, v_3) \in L^{\infty}(Q_T) \times L^{\infty}(\tilde{Q}_T) \times L^{\infty}(\tilde{Q}_T)$ (where $\tilde{Q}_T = \omega \times (0, T)$) such that $u + \varepsilon v \in \tilde{U}$ for any $\varepsilon > 0$ sufficiently small. Let (z_k, z_p) be the solution to

$$\left\{ \begin{aligned} \frac{\partial z_k}{\partial t} = & d_1 \Delta z_k + \frac{A}{1 + (p^u)^2} \left[f'((1 - \mathbb{I}_{\omega} \tau) k^u) (1 - \mathbb{I}_{\omega} \tau) z_k \right. \\ & - \left. f'((1 - \mathbb{I}_{\omega} \tau) k^u) \mathbb{I}_{\omega} k^u v_2 \right] - \delta_1 z_k - c z_k - v_1 k^u \\ & - \frac{2A p^u z_p}{(1 + (p^u)^2)^2} f'((1 - \mathbb{I}_{\omega} \tau) k^u), \\ \frac{\partial z_p}{\partial t} = & d_2 \Delta z_p + \theta \int_{\Omega} f'((1 - \mathbb{I}_{\omega}(x') \tau(x', t)) k^u(x', t)) \\ & \times [(1 - \mathbb{I}_{\omega}(x') \tau(x', t)) z_k(x', t) - \mathbb{I}_{\omega}(x') v_2(x', t) k^u(x', t)] \\ & \times \varphi(x', x) dx' \\ & - \delta_2 z_p - \mathbb{I}_{\omega} \xi z_p - \mathbb{I}_{\omega} v_3 p^u, \end{aligned} \right. \quad (10)$$

for $(x, t) \in Q_T$ together with homogeneous Neumann boundary conditions and zero initial data. One may prove in a standard way the following auxiliary result.

Lemma 1. *The following convergences hold for any $u \in \tilde{U}$*

- (i) $k^{u+\varepsilon v} \rightarrow k^u$, $p^{u+\varepsilon v} \rightarrow p^u$ in $L^2(Q_T)$, as $\varepsilon \rightarrow 0+$;
- (ii) $\frac{k^{u+\varepsilon v} - k^u}{\varepsilon} \rightarrow z_k$, $\frac{p^{u+\varepsilon v} - p^u}{\varepsilon} \rightarrow z_p$ in $L^2(Q_T)$, as $\varepsilon \rightarrow 0+$.

One may compute now the directional derivative of I :

Theorem 1.

$$dI(u)(v) = \int_{Q_T} k^u(x, t) [v_1(x, t)(1 - \lambda_k(x, t)) - \mathbb{I}_\omega(x)v_2(x, t)\beta^*(x, t)] dx dt \\ - \int_{Q_T} p^u(x, t)v_3(x, t)\mathbb{I}_\omega(x)(\beta_2 + \lambda_p(x, t)) dx dt,$$

where

$$\beta^*(x, t) = \beta_1$$

$$+ f'((1 - \mathbb{I}_\omega(x)\tau(x, t))k^u(x, t)) \left(\frac{A(x, t)\lambda_k(x, t)}{1 + p^u(x, t)^2} + \theta \int_{\Omega} \lambda_p(x', t)\varphi(x, x') dx' \right).$$

Sketch of the proof. For any $\varepsilon > 0$ sufficiently small we have that

$$I(u + \varepsilon v) - I(u) = \int_{Q_T} [(c + \varepsilon v_1)k^{u+\varepsilon v} - ck^u] dx dt \\ - \int_{Q_T} \left\{ \beta_0 [p^{u+\varepsilon v} - p^u] + \beta_1 \mathbb{I}_\omega [(\tau + \varepsilon v_2)k^{u+\varepsilon v} - \tau k^u] \right\} dx dt \\ - \int_{Q_T} \left\{ \beta_2 \mathbb{I}_\omega [(\xi + \varepsilon v_3)p^{u+\varepsilon v} - \tau p^u] \right\} dx dt.$$

By dividing the above by ε and passing to the limit ($\varepsilon \rightarrow 0+$), then using (i), (ii) in the lemma we get that

$$dI(u)(v) = \int_{Q_T} [v_1 k^u + cz_k] dx dt - \beta_0 \int_{Q_T} z_p dx dt \\ - \beta_1 \int_{Q_T} \mathbb{I}_\omega [v_2 k^u + \tau z_k] dx dt - \beta_2 \int_{Q_T} \mathbb{I}_\omega [v_3 p^u + \xi z_p] dx dt. \quad (11)$$

If we multiply Equation (8) by z_k and integrate over Q_T and use (10), then we get that

$$\begin{aligned}
 & - \int_{Q_T} \left\{ -d_1 \nabla \lambda_k(x, t) \cdot \nabla z_k(x, t) \right. \\
 & + \frac{A(x, t) \lambda_k(x, t)}{1 + p^2(x, t)} [f'((1 - \mathbb{I}_\omega(x) \tau(x, t)) k(x, t)) (1 - \mathbb{I}_\omega(x) \tau(x, t)) z_k(x, t) \\
 & - f'((1 - \mathbb{I}_\omega(x) \tau(x, t)) k(x, t)) k(x, t) \mathbb{I}_\omega(x) v_2(x, t)] \\
 & - \lambda_k(x, t) (\delta_1 z_k(x, t) + c z_k(x, t) + v_1(x, t) k(x, t)) \\
 & \left. - \frac{2A(x, t) p(x, t) z_p(x, t) \lambda_k(x, t)}{(1 + p^2(x, t))^2} f'((1 - \mathbb{I}_\omega(x) \tau(x, t)) k(x, t)) \right\} dx dt \\
 & = \int_0^T \int_\Omega \left\{ d_1 \nabla \lambda_k(x, t) \cdot \nabla z_k(x, t) - c z_k(x, t) + \beta_1 \mathbb{I}_\omega(x) \tau(x, t) z_k(x, t) \right. \\
 & - \lambda_k(x, t) z_k(x, t) \left[\frac{A(x, t) (1 - \mathbb{I}_\omega(x) \tau(x, t))}{1 + p^2(x, t)} f'((1 - \mathbb{I}_\omega(x) \tau(x, t)) k(x, t)) - \delta_1 - c \right] \\
 & - z_k(x, t) (1 - \mathbb{I}_\omega(x) \tau(x, t)) \\
 & \left. \times f'((1 - \tau(x, t)) k(x, t)) \theta \int_\Omega \lambda_p(x', t) \varphi(x, x') dx' \right\} dx dt.
 \end{aligned}$$

It follows that

$$\begin{aligned}
 & \int_{Q_T} (c z_k(x, t) - \beta_1 \mathbb{I}_\omega(x) \tau(x, t) z_k(x, t)) dx dt \\
 & = - \int_{Q_T} \int_\Omega z_k(x, t) (1 - \mathbb{I}_\omega(x) \tau(x, t)) \\
 & \quad \times f'((1 - \mathbb{I}_\omega(x) \tau(x, t)) k(x, t)) \theta \lambda_p(x', t) \varphi(x, x') dx' dx dt \\
 & \quad - \int_{Q_T} \left[\frac{A(x, t) \lambda_k(x, t)}{(1 + p^2(x, t))^2} f'((1 - \mathbb{I}_\omega(x) \tau(x, t)) k(x, t)) k(x, t) \mathbb{I}_\omega(x) v_2(x, t) \right. \\
 & \quad + \lambda_k(x, t) v_1(x, t) k(x, t) \\
 & \quad \left. + \frac{2A(x, t) p(x, t) z_p(x, t) \lambda_k(x, t)}{(1 + p^2(x, t))^2} f'((1 - \mathbb{I}_\omega(x) \tau(x, t)) k(x, t)) \right] dx dt. \quad (12)
 \end{aligned}$$

If we multiply Equation (9) by z_p and integrate over Q_T and use (10), then we get that

$$\begin{aligned}
 & - \int_{Q_T} \left\{ -d_2 \nabla \lambda_p(x, t) \cdot \nabla z_p(x, t) + \lambda_p(x, t) \theta \int_{\Omega} f'((1 - \mathbb{I}_{\omega}(x')\tau(x', t))k(x', t)) \right. \\
 & \quad \times [(1 - \mathbb{I}_{\omega}(x')\tau(x', t))z_k(x', t) - k(x', t)\mathbb{I}_{\omega}(x')v_2(x', t)]\varphi(x', x) dx' \\
 & \quad - \delta_2 \lambda_p(x, t) z_p(x, t) \\
 & \quad \left. - \mathbb{I}_{\omega}(x)\xi(x, t)\lambda_p(x, t)z_p(x, t) - \mathbb{I}_{\omega}(x)v_3(x, t)p(x, t)\lambda_p(x, t) \right\} dx dt \\
 & = \int_{Q_T} \left\{ d_2 \nabla \lambda_p(x, t) \cdot \nabla z_p(x, t) + \beta_0 z_p(x, t) + \beta_2 \mathbb{I}_{\omega}(x)\xi(x)z_p(x, t) \right. \\
 & \quad + \lambda_k(x, t)z_p(x, t) \frac{2A(x, t)p(x, t)}{(1 + p^2(x, t))^2} f((1 - \mathbb{I}_{\omega}(x)\tau(x, t))k(x, t)) + \delta_2 \lambda_p(x, t)z_p(x, t) \\
 & \quad \left. + \mathbb{I}_{\omega}(x)\xi(x, t)\lambda_p(x, t)z_p(x, t) \right\} dx dt.
 \end{aligned} \tag{13}$$

It follows that

$$\begin{aligned}
 & - \int_{Q_T} \beta_0 z_p(x, t) dx dt - \beta_2 \int_{Q_T} \mathbb{I}_{\omega}(x)\xi(x, t)z_p(x, t) dx dt \\
 & = \int_{Q_T} \lambda_k(x, t)z_p(x, t) \frac{2A(x, t)p(x, t)}{(1 + p^2(x, t))^2} f((1 - \mathbb{I}_{\omega}(x)\tau(x, t))k(x, t)) dx dt \\
 & \quad + \theta \int_{Q_T} \int_{\Omega} \lambda_p(x, t) f'((1 - \mathbb{I}_{\omega}(x')\tau(x', t))k(x', t)) \\
 & \quad \times [(1 - \mathbb{I}_{\omega}(x')\tau(x', t))z_k(x', t) - k(x', t)\mathbb{I}_{\omega}(x')v_2(x', t)]\varphi(x', x) dx' dx dt \\
 & \quad - \int_{Q_T} \mathbb{I}_{\omega}(x)v_3(x, t)p(x, t)\lambda_p(x, t) dx dt.
 \end{aligned} \tag{14}$$

By (12) and (14) we get that

$$\begin{aligned}
 & \int_0^T \int_{\Omega} (cz_k(x, t) - \beta_0 z_p(x, t) - \beta_1 \mathbb{I}_{\omega}(x) \tau(x, t) z_k(x, t) \\
 & \quad - \beta_2 \mathbb{I}_{\omega}(x) \xi(x, t) z_p(x, t)) dx dt \\
 & = \int_{Q_T} \left[- \frac{A(x, t) \lambda_k(x, t)}{1 + p^2(x, t)} f'((1 - \mathbb{I}_{\omega}(x) \tau(x, t)) k(x, t)) k(x, t) \mathbb{I}_{\omega}(x) v_2(x, t) \right. \\
 & \quad \left. - v_1(x, t) k(x, t) \lambda_k(x, t) \right] dx dt \\
 & \quad - \theta \int_{Q_T} \int_{\Omega} \lambda_p(x, t) f'((1 - \mathbb{I}_{\omega}(x') \tau(x', t)) k(x', t)) k(x', t) \\
 & \quad \times \mathbb{I}_{\omega}(x') v_2(x', t) \varphi(x', x) dx' dx dt - \int_{Q_T} \mathbb{I}_{\omega}(x) v_3(x, t) p(x, t) \lambda_p(x, t) dx dt. \quad (15)
 \end{aligned}$$

By (11) and (15) we get that

$$\begin{aligned}
 dI(u)(v) & = \int_{Q_T} k(x, t) [v_1(x, t) (1 - \lambda_k(x, t)) - v_2(x, t) \mathbb{I}_{\omega}(x) \beta^*(x, t)] dx dt \\
 & \quad - \int_{Q_T} p(x, t) v_3(x, t) \mathbb{I}_{\omega}(x) (\beta_2 + \lambda_p(x, t)) dx dt,
 \end{aligned}$$

and so we get the conclusion of the theorem.

In Paragraph 3.1 we describe the conceptual gradient-type algorithm for the optimal control procedure. In Section 4 we describe the numerical methods adopted for the solutions of System (4) and (8)-(9) with the above mentioned boundary and initial/final conditions. In Section 5 we report and discuss the computational results.

3.1. Conceptual algorithm

STEP 0: Set $iter := 0$, Choose $\maxiter, \delta > 0$.

STEP 1: Initialize $(c^{(iter)}, \tau^{(iter)}, \xi^{(iter)}) \in \tilde{U}$.

STEP 2: Compute $k^{(iter)}, p^{(iter)}, \lambda_k^{(iter)}, \lambda_p^{(iter)}, (\beta^*)^{(iter)}$ corresponding to $(c^{(iter)}, \tau^{(iter)}, \xi^{(iter)})$, and $I^{(iter)} = I(c^{(iter)}, \tau^{(iter)}, \xi^{(iter)})$.

- STEP 3: Choose $(v_1, v_2, v_3) \in L^\infty(Q_T) \times L^\infty(\tilde{Q}_T) \times L^\infty(\tilde{Q}_T)$ such that $(c^{(iter)}, \tau^{(iter)}, \xi^{(iter)}) + \varepsilon_0(v_1, v_2, v_3) \in U$ ($\varepsilon_0 \geq 0$) and that $v_1(1 - \lambda_k^{(iter)}) \geq 0$, $-v_2 \mathbb{I}_\omega(\beta^*)^{(iter)} \geq 0$, $-v_3 \mathbb{I}_\omega(\beta_2 + \lambda_p^{(iter)}) \geq 0$ a.e. $(x, t) \in Q_T$.
- STEP 4: Compute $(c^{(new)}, \tau^{(new)}, \xi^{(new)}) := (c^{(iter)}, \tau^{(iter)}, \xi^{(iter)}) + \varepsilon_0(v_1, v_2, v_3)$.
- STEP 5: Compute η_0 which maximizes $I(\eta(c^{(iter)}, \tau^{(iter)}, \xi^{(iter)}) + (1-\eta)(c^{(new)}, \tau^{(new)}, \xi^{(new)}))$, for $\eta \in [0, 1]$. Set $(c^{(iter+1)}, \tau^{(iter+1)}, \xi^{(iter+1)}) = \eta_0(c^{(iter)}, \tau^{(iter)}, \xi^{(iter)}) + (1 - \eta_0)(c^{(new)}, \tau^{(new)}, \xi^{(new)})$.
- STEP 6: If $I^{(iter+1)} - I^{(iter)} < \delta$, then $(c^*, \tau^*, \xi^*) := (c^{(iter)}, \tau^{(iter)}, \xi^{(iter)})$ else if $iter > maxiter$, then $(c^*, \tau^*, \xi^*) := (c^{(iter)}, \tau^{(iter)}, \xi^{(iter)})$ else $iter := iter + 1$ and GO TO STEP 3.

In the numerical tests, we will use:

$$v_1(x, t) \in \text{sgn}(1 - \lambda_k^{(iter)}(x, t)),$$

$$v_2(x, t) \in -\text{sgn}(\beta^*(x, t))^{(iter)}, \quad v_3(x, t) \in -\text{sgn}(\beta_2 + \lambda_p^{(iter)}(x, t)),$$

a.e. $(x, t) \in Q_T$, and a.e. $(x, t) \in \tilde{Q}_T$, respectively.

4. Numerical methods

Equations (4) and (8)–(9) are discretized by finite elements in space and finite differences in time.

4.1. Space discretization

We first apply a standard Galerkin procedure to the weak formulations of systems (4) and (8)–(9), with the associated boundary and initial/ final conditions, respectively. The two-dimensional set $\bar{\Omega} = [-1, 1]^2$ is discretized by a uniform grid of 64×64 bilinear finite elements ($\mathbf{Q}^1$), yielding a total amount of $N = 4225$ discretization nodes. The stiffness matrix is computed exactly, whereas the mass matrix is obtained by applying the mass lumping technique.

In the following, we describe in detail this procedure applied to System (4), being the treatment of equations (8)–(9) analogous.

Let us first denote by V the Sobolev space $H^1(\Omega)$, by $(\cdot, \cdot)$ the L^2 - inner product and by $a_1(\cdot, \cdot)$, $a_2(\cdot, \cdot) : V \times V \rightarrow \mathbb{R}$ the bilinear forms

$$\begin{aligned} a_1(u, v) &:= \int_{\Omega} d_1 \nabla u \cdot \nabla v \, dx, & \text{for all } u, v \in V, \\ a_2(u, v) &:= \int_{\Omega} d_2 \nabla u \cdot \nabla v \, dx, & \text{for all } u, v \in V. \end{aligned}$$

Then, using Green's formula, the variational formulation of problem (4) reads: given $(c, \tau) \in U$, find k, p such that

$$\begin{cases} \frac{\partial}{\partial t}(k(t), \phi) + a_1(k(t), \phi) = (G_1(k(t), p(t), c(t), \tau(t)), \phi), & \text{for all } \phi \in V, \\ \frac{\partial}{\partial t}(p(t), \phi) + a_2(p(t), \phi) = (G_2(k(t), p(t), c(t), \tau(t)), \phi), & \text{for all } \phi \in V, \end{cases} \quad (16)$$

subject to the initial conditions on k and p , and where G_1 and G_2 are the reaction terms in (4).

Let us now introduce the $\mathbf{P}^1$ finite element space V^h , associated to a partition of the one-dimensional domain Ω into $N - 1$ subintervals of size h .

The space semi-discretization of problem (16) then reads:

find $k_h, p_h : [0, T] \rightarrow V^h$ such that

$$\begin{cases} \frac{\partial}{\partial t}(k_h(t), \phi_h) + a_1(k_h(t), \phi_h) = (G_1(k_h(t), p_h(t), c_h(t), \tau_h(t)), \phi_h), \\ \frac{\partial}{\partial t}(p_h(t), \phi_h) + a_2(p_h(t), \phi_h) = (G_2(k_h(t), p_h(t), c_h(t), \tau_h(t)), \phi_h), \end{cases}$$

for all $\phi_h \in V^h$, where $c_h, \tau_h : [0, T] \rightarrow V^h$ are convenient finite element approximations of c and τ . The previous variational equation is equivalent to

$$\begin{cases} \frac{\partial}{\partial t}(k_h(t), \phi_i) + a_1(k_h(t), \phi_i) = (G_1(k_h(t), p_h(t), c_h(t), \tau_h(t)), \phi_i), \\ \frac{\partial}{\partial t}(p_h(t), \phi_i) + a_2(p_h(t), \phi_i) = (G_2(k_h(t), p_h(t), c_h(t), \tau_h(t)), \phi_i), \end{cases} \quad (17)$$

for all $i = 1, \dots, N$, where $\phi_i, i = 1, \dots, N$, are the *hat basis functions* of the $\mathbf{P}^1$ finite element space V^h .

Let us remind that hat basis functions are finite element functions (thus piecewise bilinear) that are equal to one at a given mesh node and decrease linearly to zero at neighboring points.

Expanding the unknowns $k_h(x, t)$, $p_h(x, t)$ and $c_h(x, t)$, $\tau_h(x, t)$ with respect to the basis $\{\phi_i\}_{i=1}^N$, we have

$$\begin{aligned} k_h(x, t) &= \sum_{j=1}^N K_j(t) \phi_j(x), & p_h(x, t) &= \sum_{j=1}^N P_j(t), \\ c_h(x, t) &= \sum_{j=1}^N C_j(t) \phi_j(x), & \tau_h(x, t) &= \sum_{j=1}^N T_j(t), \end{aligned} \quad (18)$$

where $K_j, P_j: [0, T] \rightarrow \mathbb{R}$ are unknown scalar functions to be determined, and we assume that $C_j, T_j: [0, T] \rightarrow \mathbb{R}$ are given.

Denoting by A_1, A_2 the stiffness matrices associated with the bilinear forms $a_1(\cdot, \cdot), a_2(\cdot, \cdot)$, by M the lumped mass matrix, and substituting the expressions (18) into system (17), the semi-discrete problem then reads:

find $\mathbf{K}, \mathbf{P}: [0, T] \rightarrow \mathbb{R}^N$ such that

$$\begin{cases} M \frac{d\mathbf{K}}{dt}(t) + A_1 \mathbf{K}(t) = M \mathbf{G}_1(\mathbf{K}(t), \mathbf{P}(t), \mathbf{C}(t), \mathbf{T}(t)), \\ M \frac{d\mathbf{P}}{dt}(t) + A_2 \mathbf{P}(t) = M \mathbf{G}_2(\mathbf{K}(t), \mathbf{P}(t), \mathbf{C}(t), \mathbf{T}(t)), \end{cases} \quad (19)$$

where

$$\begin{aligned} \mathbf{K}(t) &= [\dots, K_j(t), \dots]^T, \\ \mathbf{P}(t) &= [\dots, P_j(t), \dots]^T, \\ \mathbf{C}(t) &= [\dots, C_j(t), \dots]^T, \\ \mathbf{T}(t) &= [\dots, T_j(t), \dots]^T, \\ \mathbf{G}_1(\mathbf{K}(t), \mathbf{P}(t), \mathbf{C}(t), \mathbf{T}(t)) &= [\dots, G_1(K_j(t), P_j(t), C_j(t), T_j(t)), \dots]^T, \\ \mathbf{G}_2(\mathbf{K}(t), \mathbf{P}(t), \mathbf{C}(t), \mathbf{T}(t)) &= [\dots, G_2(K_j(t), P_j(t), C_j(t), T_j(t)), \dots]^T. \end{aligned}$$

4.2. Time discretization

After space discretization by finite elements, we obtain the semi-discrete problem (19) that consists of a system of ODEs. We solve this system by employing a first order semi-implicit finite difference scheme, where the linear diffusion and reaction terms are approximated by Backward Euler, whereas the non-linear reaction terms are approximated by Forward Euler.

In formulae, denoting by Δt the time step size, at the generic time step n , given $\mathbf{K}^n, \mathbf{P}^n$, we have to find $\mathbf{K}^{n+1}, \mathbf{P}^{n+1}$ such that

$$\begin{cases} M \frac{\mathbf{K}^{n+1} - \mathbf{K}^n}{\Delta t} + A_1 \mathbf{K}^{n+1} = M \mathbf{G}_1(\mathbf{K}^n, \mathbf{P}^n, \mathbf{C}^n, \mathbf{T}^n), \\ M \frac{\mathbf{P}^{n+1} - \mathbf{P}^n}{\Delta t} + A_2 \mathbf{P}^{n+1} = M \mathbf{G}_2(\mathbf{K}^n, \mathbf{P}^n, \mathbf{C}^n, \mathbf{T}^n). \end{cases}$$

As a result, at each time step it is required the solution of the linear system of algebraic equations

$$\begin{bmatrix} \frac{1}{\Delta t} M + A_1 & 0 \\ 0 & \frac{1}{\Delta t} M + A_2 \end{bmatrix} \begin{bmatrix} \mathbf{K}^{n+1} \\ \mathbf{P}^{n+1} \end{bmatrix} = \begin{bmatrix} \frac{1}{\Delta t} M \mathbf{K}^n + M \mathbf{G}_1(\mathbf{K}^n, \mathbf{P}^n, \mathbf{C}^n, \mathbf{T}^n) \\ \frac{1}{\Delta t} M \mathbf{P}^n + M \mathbf{G}_2(\mathbf{K}^n, \mathbf{P}^n, \mathbf{C}^n, \mathbf{T}^n) \end{bmatrix},$$

which in our specific case has the dimension of 8450 degrees of freedom. The linear system is solved by the Gaussian elimination with the built-in function of MATLAB. The time step size is $\Delta t = 0.05$.

5. Numerical simulations

Let us use the above algorithm with the following parameters' values $d_1 = 1$, $d_2 = 1$, $\delta_1 = 0.05$, $\delta_2 = 0.03$, $A = 1$, $s = 0.6$, $\theta = 2$, $\alpha_1 = 0.7$, $\alpha_2 = 1$, $\gamma = 4$, $\varphi = 0.3$, and initial conditions

$$k_0(x_1, x_2) = 0.1 \exp \left\{ \frac{-(x_1 - 0.5)^2 - (x_2 - 0.5)^2}{0.1} \right\}$$

and

$$p_0(x_1, x_2) = \exp\{x_1 + x_2\};$$

see Fig. 1. We have taken $\beta_0 = \beta_1 = \beta_2 = 1$ and $L = 0.5$.

We consider the following four scenarios; as time horizon for the control we have taken $T = 5$:

- **case 0**: no control is applied;
- **case 1**: the control region ω is the small circle displayed in Fig. 2 (left);

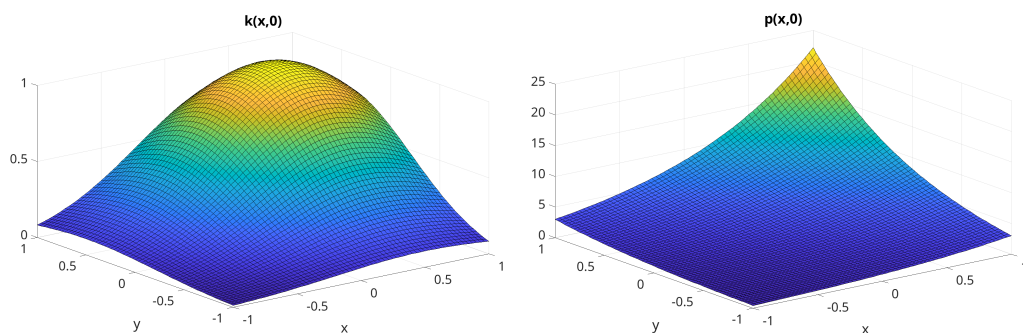


Figure 1: Initial distributions (at $t = 0$) of state variables $k(x, t)$ (left) and $p(x, t)$ (right)

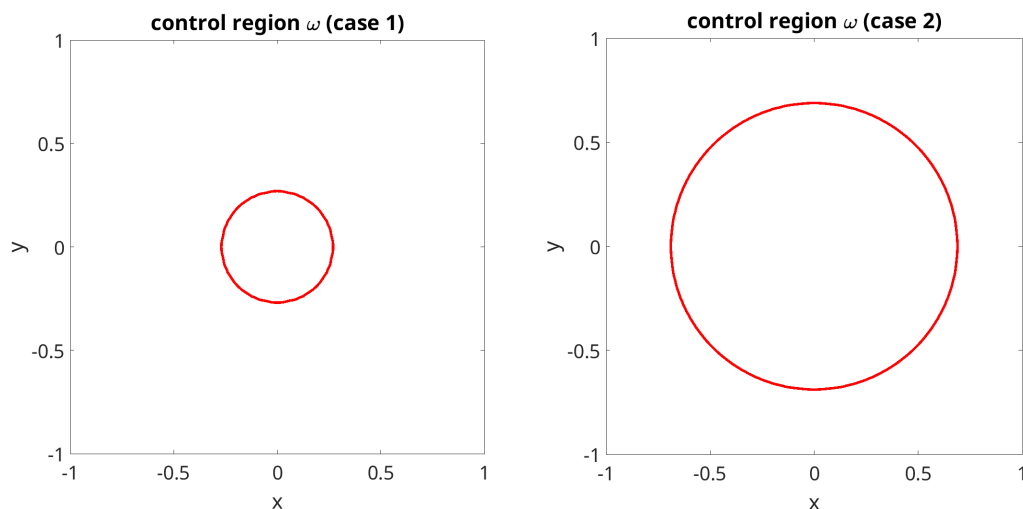


Figure 2: Control regions ω : small (case 1, left) and big (case 2, right)

- **case 2:** the control region ω is the big circle displayed in Fig. 2 (right);
- **case 3:** the control region ω is the whole domain $\Omega = [-1, 1]^2$.

The numerical results/tests agree with what intuition tells us. However, besides of what agrees with the intuition, we get additional information. Note that the numerical results give also an evaluation of the optimal control magnitude (c^*, τ^*, ξ^*) and of the corresponding states $k^*(x, t), p^*(x, t)$.

Figure 4. Since our main goal is to get a maximal value of the cost functional it is natural noticing a decrease of the capital when time increases (because there is no interest in keeping a high reserve of k for time larger than the time horizon T).

Figure 5. The level of the optimal consumption per unit of capital, c^* , becomes maximal when time is close to the horizon T . The optimal environmental taxation

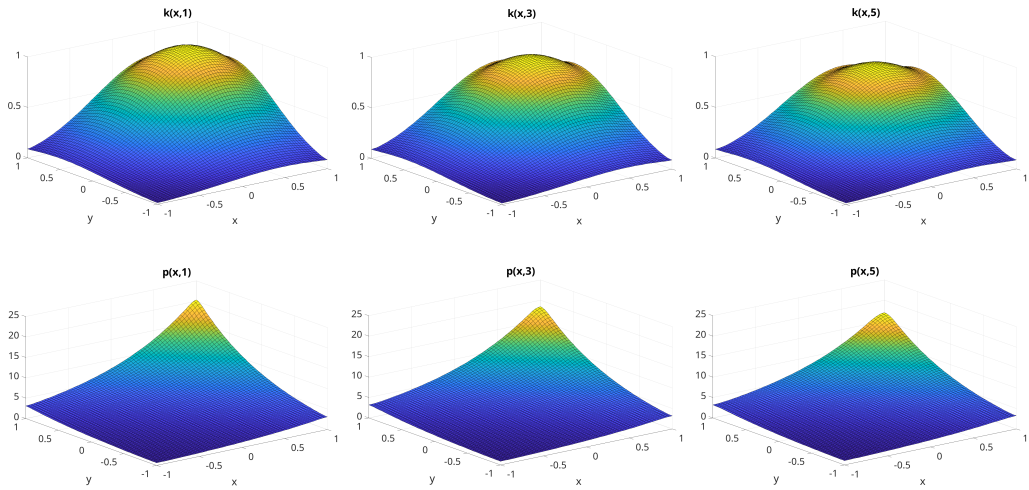


Figure 3: **Case 0**, no control. Space distributions of state variables $k(x, t)$ (first row) and $p(x, t)$ (second row) at three time instants

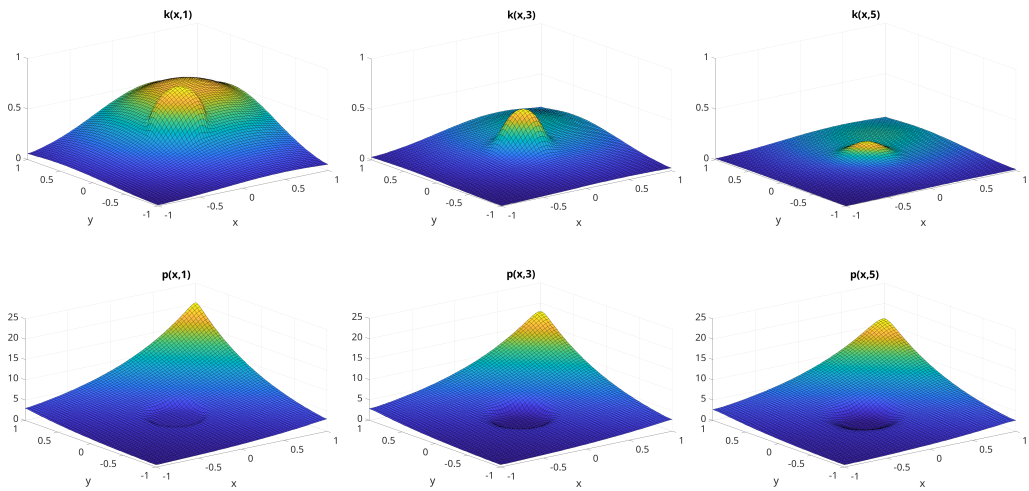


Figure 4: **Case 1**, small control region. Space distributions of the optimal state variables $k(x, t)$ (first row) and $p(x, t)$ (second row) at three time instants

rate and the optimal pollution abatement approach 0 as time approaches the time horizon T . Note that $c^* + \tau^* = 1 - s$ for any $x \in \omega$ and $t \in (0, T)$.

The same remarks are true for the other cases.

Moreover, in Case 2 (the large control region) we get a higher level of capital and a lower level of pollution.

Figure 8 (corresponding to Case 3) shows an important diminishment of $k(x, t)$ as $t \rightarrow T$. On the other hand, Figure 9 shows that in this case $\tau^*(x, t)$ is

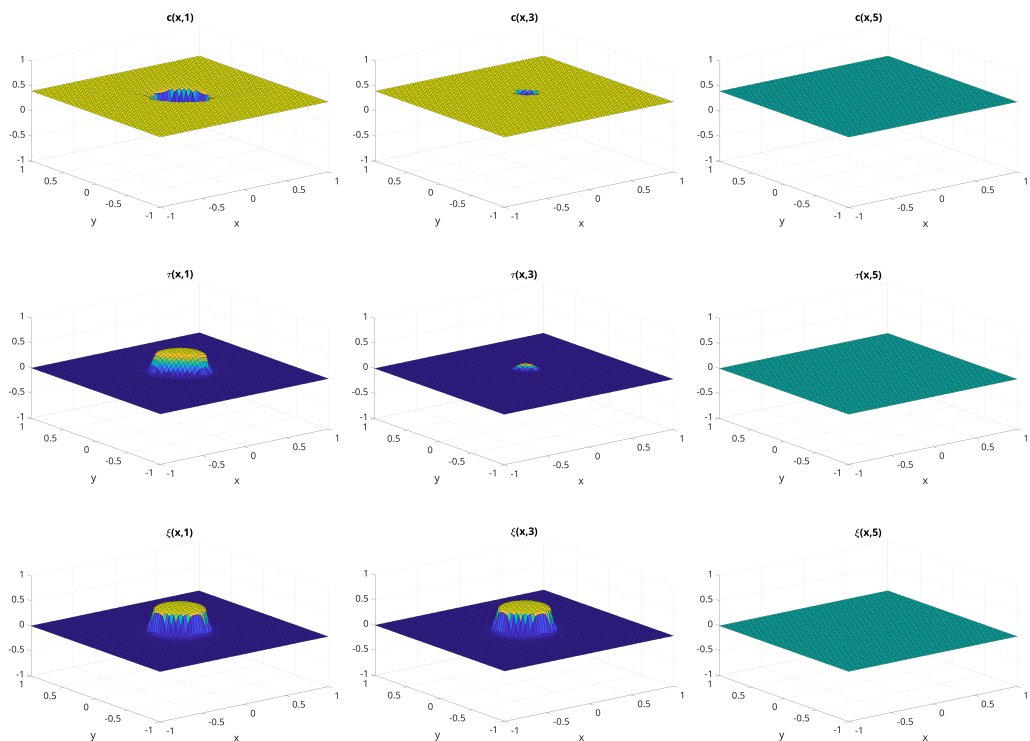


Figure 5: **Case 1**, small control region. Space distributions of the optimal control variables $c(x, t)$ (first row), $\tau(x, t)$ (second row) and $\xi(x, t)$ (third row) at three time instants

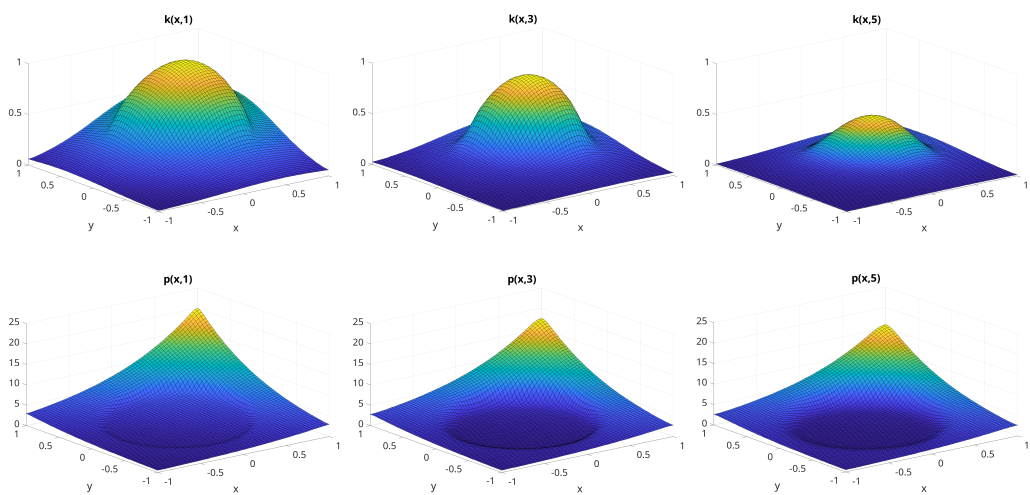


Figure 6: **Case 2**, big control region. Space distributions of the optimal state variables $k(x, t)$ (first row) and $p(x, t)$ (second row) at three time instants

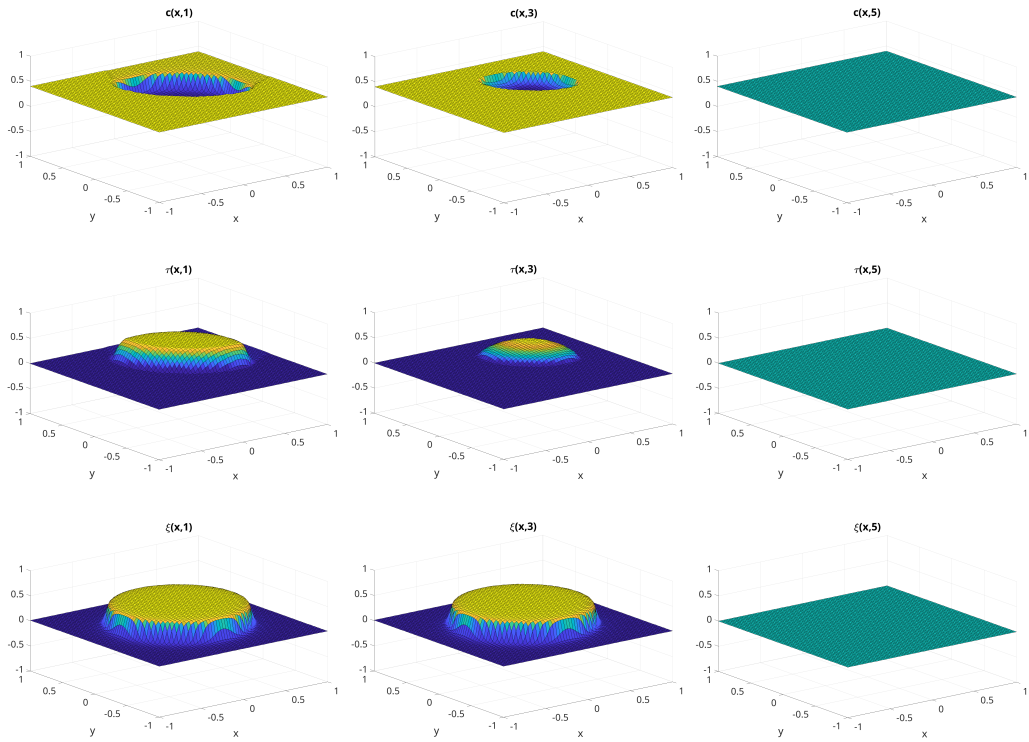


Figure 7: **Case 2**, big control region. Space distributions of the optimal control variables $c(x, t)$ (first row), $\tau(x, t)$ (second row) and $\xi(x, t)$ (third row) at three time instants

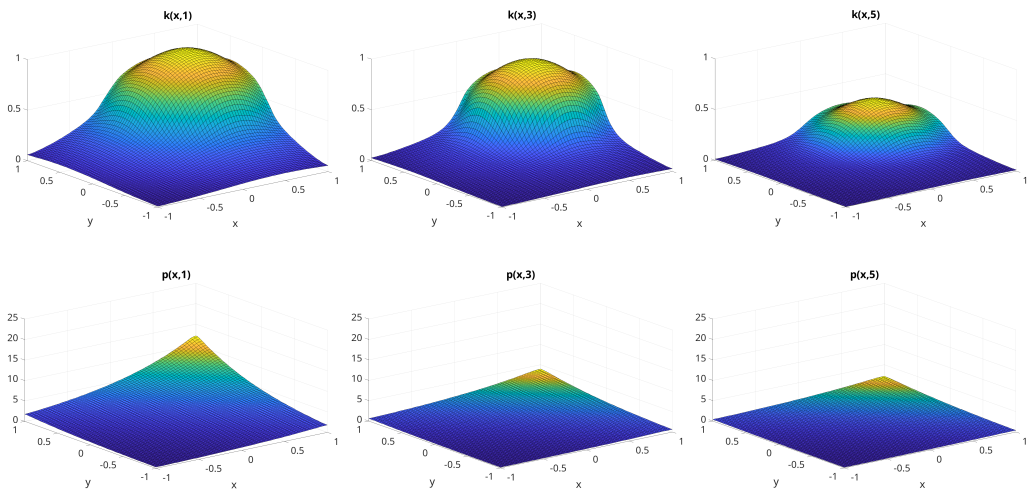


Figure 8: **Case 3**, the control region is the whole domain. Space distributions of the optimal state variables $k(x, t)$ (first row) and $p(x, t)$ (second row) at three time instants

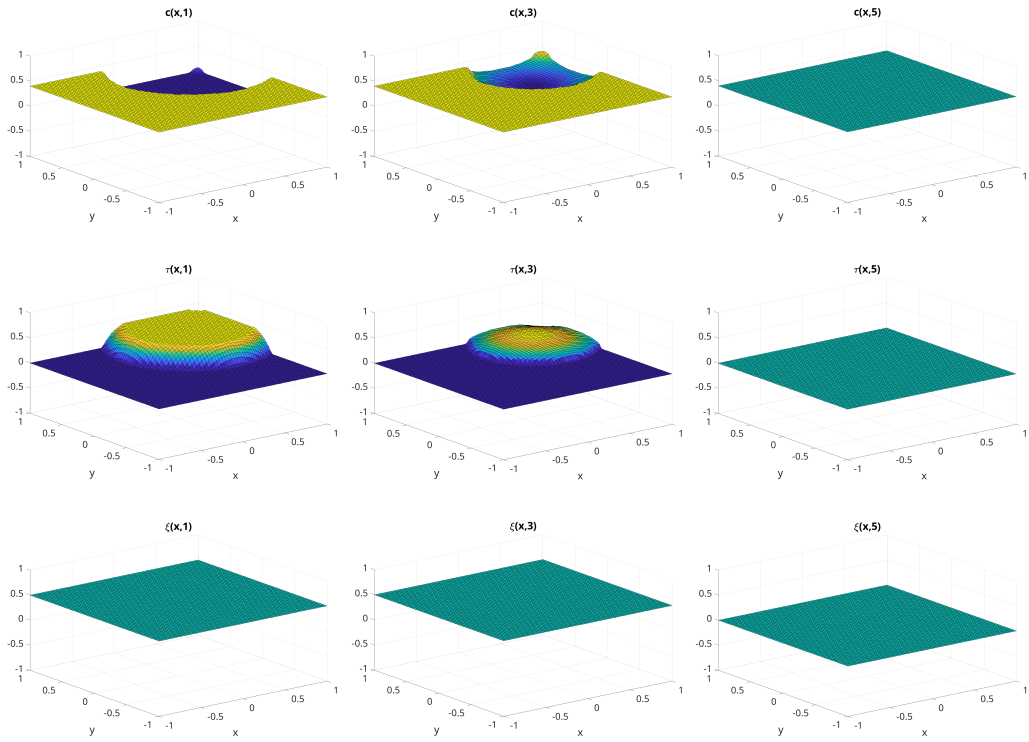


Figure 9: **Case 3**, the control region is the whole domain. Space distributions of the optimal control variables $c(x, t)$ (first row), $\tau(x, t)$ (second row) and $\xi(x, t)$ (third row) at three time instants

big for x close to the region with initial high level of pollution if t is small and it is small for large t or for x far from this region.

6. Concluding remarks

The study presented a spatially structured economic growth model that incorporates pollution dynamics, environmental taxation, and localized intervention strategies. By formulating and analyzing an optimal control problem with multiple interacting objectives – maximizing intertemporal utility, minimizing pollution, and accounting for both taxation disutility and intervention costs – we have extended previous frameworks in environmental economics and control theory to include the spatial localization of environmental policies. Numerical simulations evidence that restricting pollution control actions to optimally chosen subregions can effectively reduce pollution diffusion in the whole eco-system while maintaining a satisfactory level of production. The results emphasize the importance of spatial heterogeneity in designing efficient environmental policies: targeted

interventions, rather than uniform controls, yield better trade-offs between economic growth and ecological sustainability. Beyond its computational findings, the proposed framework highlights the potential of regional control theory as a valuable tool for eco-economic systems. Future research will explore more complex spatial domains, dynamic intervention costs, and stochastic factors affecting both production and pollution dynamics, to further support evidence-based environmental management in geographically distributed economic systems.

Credit author statement. All authors have contributed in an equal manner to the development of the research project reported in this manuscript. In addition, Simone Scacchi has been responsible for the numerical software.

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